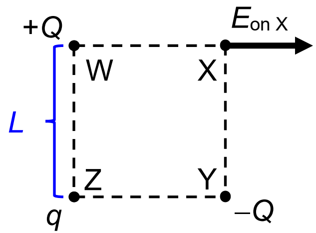
19 **D** electric force is vector quantity, and the charge at X is negative so electric field strength actually points to right:



$$\overrightarrow{E_X} = \overrightarrow{E_{\text{by W}}} + \overrightarrow{E_{\text{by Z}}} + \overrightarrow{E_{\text{by Y}}}$$

vertically :

$$0 = \frac{q}{L^2 + L^2}\cos 45^\circ - \frac{Q}{L^2}$$

$$= \frac{q}{L^2 + L^2}\cos 45^\circ - \frac{Q}{L^2}$$

$$= \frac{q}{2L^2}\frac{1}{\sqrt{2}} - \frac{Q}{L^2}$$

$$q = \frac{2\sqrt{2}\,\cancel{L^2}}{\cancel{L^2}}Q$$

20 **C** resistance of R alone is larger than the effective resistance of R in parallel with R